Exercise 4.6

1	2	3	4	5	6	7	8	9	10
(3)	(2)	(3)	(1)	(2)	(1)	(3)	(1)	(4)	(4)
11	12	13	14	15	16	17	18	19	20
(3)	(2)	(2)	(1)	(3)	(3)	(2)	(2)	(4)	(4)

Exercise 5.1

1. $x^2+y^2\pm 10y=0$ 2. $(x-2)^2+(y+1)^2=50$
3. $x^2+y^2+4x+4y+4=0$ or $x^2+y^2+20x+20y+100=0$ 4. $x^2+y^2-4x-6y-12=0$
5. $x^2+y^2-5x+3y-22=0$ 6. $x^2+y^2=1$
7. $x^2+y^2-6x-4y+4=0$ 8. ± 12
9. $x-5y+8=0,\ 5x+y-12=0$ 10. out side, inside, outside
11. (i) $(0,-2),0$ (ii) $(-3,2),3$ (iii) $\left(\frac{1}{2},-1\right),\frac{\sqrt{17}}{2}$ (iv) $\left(\frac{3}{2},-1\right),\frac{3}{2}$
12. $p=q=3,\ (1,0),5$

Exercise 5.2

1. (i) $y^2=16x$ (ii) $3x^2=-4y$ (iii) $(y+2)^2=12(x-1)$ (iv) $y^2=16x$

2. (i) $\frac{x^2}{36}+\frac{y^2}{27}=1$ (ii) $\frac{x^2}{9}+\frac{y^2}{25}=1$ (iii) $\frac{16x^2}{625}+\frac{y^2}{25}=1$ (iii) $\frac{x^2}{8}+\frac{y^2}{16}=1$

3. (i) $\frac{9x^2}{16}-\frac{9y^2}{20}=1$ (ii) $\frac{(x-2)^2}{12}-\frac{(y-1)^2}{24}=1$ (iii) $\frac{x^2}{16}-\frac{9y^2}{64}=1$

4.

	Vertex	Focus	Equation of directrix	Length of latus rectum
i.	$(0,0)$	$(4,0)$	$x=-4$	16
ii.	$(0,0)$	$(0,6)$	$y=-6$	24
iii.	$(0,0)$	$(-2,0)$	$x=2$	8
iv.	$(1,-2)$	$(1,-4)$	$y=0$	8
v.	$(1,2)$	$(3,2)$	$x=-1$	8

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